CIRC HQ STUDENT HANDOUT | GRADE 5

Cargo Sorter

Goal: Design a mechanical sorting system that routes model cargo by one measurable property.

35 minutes | 8 steps | Engineering and Design, Computer Science and Programming | Playbook A meeting 15, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Report batch accuracy and show the revision that prevented the most common error.

You will use

- Mixed model cargo
- Cardboard
- Ramps
- Cups
- Craft sticks
- Tape
- Optional magnets

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place mixed cargo in the starting cup.
- Choose two ending bins.

2 Step 2: Sorter safety — 3 min

- Keep fingers outside every moving path.
- Keep magnets away from electronics.
- Feed one cargo item at a time.

3 Step 3: Plan the sort — 4 min

- Choose one sorting rule.
- Draw the travel path.
- Label both ending bins.

4 Step 4: Build the sorter — 9 min

- Build the starting ramp.
- Add one part that sends pieces to a bin.
- Place the bins under the exits.

5 Step 5: Run the batch — 7 min

- Feed one mixed item.
- Record its ending bin.
- Repeat for all test pieces.

6 Step 6: Fix the errors — 4 min

- Change the sorting part.
- Test all pieces again.

7 Step 7: Reset the cargo — 2 min

- Count every cargo item.
- Return magnets to the marked box.

8 Step 8: Exit check — 3 min

- Write the final error count.
- Name the sorting rule.

What to turn in

Batch accuracy and the fix that stopped the most common error.

No computer? Another way to do it

Sort by hand with a written rule first.

Finished early? Next challenge

Third Destination: Add a third output without reducing the accuracy of the original two groups.

Stay safe

Keep magnets away from personal electronics and feed cargo without putting fingers inside a moving path.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.